

NeuroRest AI: Intelligent Sleep Apnea and Stress Risk Analyzer

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Abstract

This project presents an end-to-end Artificial Intelligence system designed to analyze sleep health and predict stress levels. By replacing traditional Random Forest models with a PyTorch-based Multi-Task Neural Network, the system achieves simultaneous classification of sleep disorders and regression of stress levels. Key technical improvements include the total elimination of data leakage, the deployment of the model via a Gradio interface, and the implementation of Batch Normalization and Dropout for enhanced training stability.

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1 Introduction

Sleep disorders and elevated stress levels are significant health challenges in modern society. **NeuroRest AI** is a diagnostic tool designed to analyze user vitals and lifestyle data—such as blood pressure, heart rate, and physical activity—to provide actionable health insights. The system utilizes deep learning to categorize health risks and offer personalized coaching tips through a cloud-hosted interface.

Project URL: <https://huggingface.co/spaces/ZohaSarwar/NeuroRest-AI>

2 Data Description and Preprocessing

The model was trained using the *Sleep Health and Lifestyle Dataset*, which contains 374 observations of individual health metrics.

2.1 Preprocessing and Feature Engineering

To ensure the model performed accurately, the following preprocessing steps were implemented:

- **Categorical Encoding:** Variables such as *Gender*, *Occupation*, and *BMI Category* were converted into numerical formats using label encoding and manual mapping to ensure compatibility with the neural network.
- **Blood Pressure Refinement:** The composite "Blood Pressure" string was split into two distinct numerical features: *BP_Systolic* and *BP_Diastolic*.
- **Feature Scaling:** All 11 input features were normalized using a `StandardScaler`. This prevents features with larger numerical ranges from disproportionately influencing the model's weights.

3 Methodology and Architecture

3.1 System Architecture Overview

The system follows a modular architecture consisting of the data pipeline, the neural network engine, and the interface layer:

Layer	Component	Function
Data Layer	CSV Dataset	Stores lifestyle observations.
Preprocessing	StandardScaler	Normalizes inputs for neural stability.
Engine	Multi-Task Net	Shared backbone with dual-head output.
Interface	Gradio (app.py)	Renders dynamic HTML health cards.

3.2 Multi-Task Neural Network Design

NeuroRest AI utilizes a Multi-Task Learning (MTL) architecture built in PyTorch. This design allows for a shared representation of user health before branching into specific predictive outputs:

- **Shared Backbone:** Features three fully connected linear layers (128, 64, and 32 units). It includes ReLU activation functions, Batch Normalization for training stability, and 25% Dropout to prevent overfitting.
- **Classification Head:** A dedicated output layer that categorizes users into one of three classes: None, Insomnia, or Sleep Apnea.
- **Regression Head:** A dedicated output layer that predicts a continuous Stress Level on a scale of 3–8.

3.3 Model Specification Table

Specification	Detail
Model Type	PyTorch Multi-Task Neural Network
Input Dimensions	11 Features (Stress Level removed from input)
Hidden Architecture	Linear(128) → Linear(64) → Linear(32)
Regularization	Dropout (0.25) & Batch Normalization
Optimizers	AdamW + CosineAnnealingLR Scheduler
Loss Weights	1.0 (Classification) + 0.5 (Regression)

4 Challenges Faced During Implementation

4.1 Data Leakage Prevention

A major technical improvement was the removal of "Stress Level" from the input feature set. In previous versions, using the target variable as an input led to inflated performance metrics. The current model learns to predict stress purely from physiological and behavioral data.

4.2 Training Convergence and Normalization

Early versions struggled with training stability due to the varying scales of input features like Age and Daily Steps. Implementation of `StandardScaler` and `BatchNorm1d` layers was essential to prevent vanishing gradients, allowing the model to converge smoothly.

4.3 Multi-Task Loss Balancing

Balancing the two loss functions (Cross-Entropy and MSE) required calibration. We found that assigning a weight of 1.0 to classification and 0.5 to regression provided the best overall performance without one task dominating the learning process.

5 Implementation and Deployment

5.1 Gradio User Interface

The front-end application was developed using the Gradio library. It provides a responsive web interface where users can input data via sliders and dropdown menus. The backend uses custom "Wrapper" classes to handle the PyTorch inference and generate visual coaching cards in HTML.

5.2 Cloud Hosting

The application is hosted on Hugging Face Spaces. The deployment repository consists of the application logic (`app.py`), the trained model weights (`model.pth`), and the feature encoders (`encoders.pkl`).

6 Results

The Multi-Task model demonstrated strong performance on the validation set:

- **Sleep Disorder Classification Accuracy:** 66.67%
- **Stress Level Regression (R^2 Score):** 0.8286
- **Stress Level Regression (RMSE):** 0.7331

7 Conclusion

NeuroRest AI demonstrates the effectiveness of Multi-Task Deep Learning in the health-tech domain. By combining sophisticated neural network architectures with an accessible cloud-based interface, the project provides a scalable framework for personal health monitoring and risk detection.

8 References

1. *NeuroRest AI: Technical Project Specifications*, 2026.
2. *Sleep Health and Lifestyle Dataset*, Public Research Documentation.
3. *Deep Learning Upgrade Changelog*, Internal Documentation.